SILIGER



CX4170a

17" DEEP 4U CHASSIS

ENGLISH

In this guide

This guide covers chassis access, component installation, and reassembly.

Choose your cooling configuration and check component clearances before starting assembly. Supported components may compete for the same space.

Before installation: Follow the installation instructions supplied with your motherboard, power supply, graphics card, and cooler.

Maintenance: Before opening a chassis that already has components installed, switch the system off and unplug it.

Fasteners: Tighten chassis fasteners by hand. Do not force a screw that does not thread correctly.

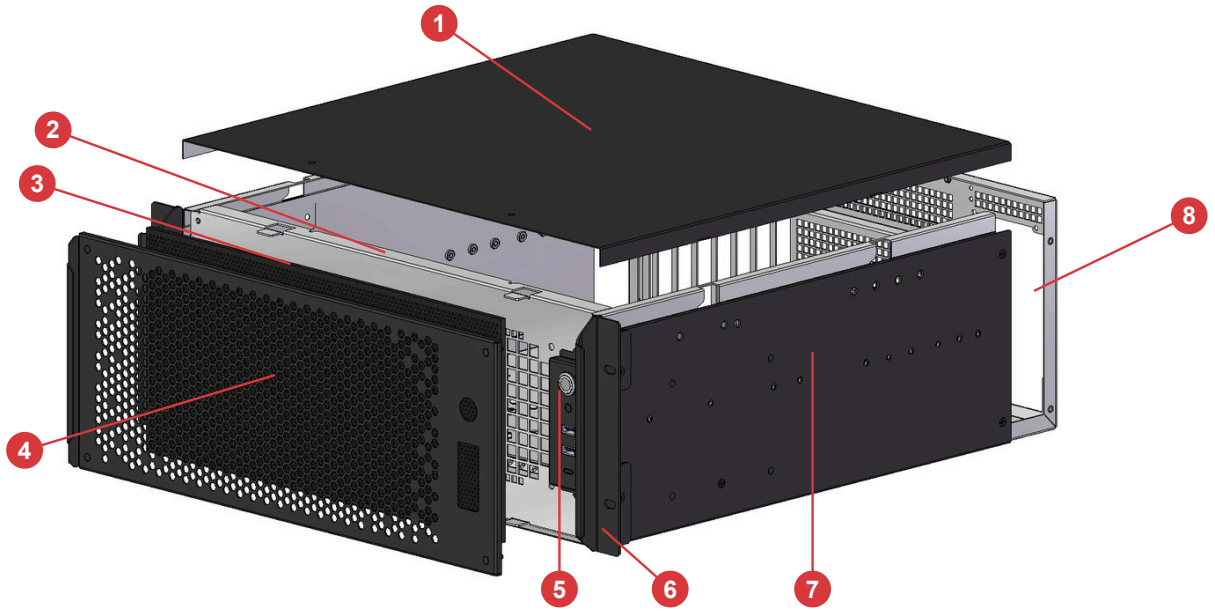
Exploded assembly & part key 2
Clearance drawing — drive cage installed 3
Clearance drawing — cage removed 4
Front-view component clearances 5
Dimensions & compatibility 6
Hardware kit 7
Front ports & connections 8
Remove the lid 9
Remove the faceplate & filter 10
Remove the motherboard tray 11
Install rack ears & handles 12
Install front fans 13
Install a front AIO radiator 14
Radiator & drive-cage clearance 15
Install push-pull cooling 16
Remove the internal drive cage 17
Install drives in the cage 18
Drive-cage layout examples 19
Prepare the motherboard tray 20
Install the motherboard 21
Install the power supply 22
Install a GPU or PCIe card 23
Install a 2.5-inch SSD 24
Install the rear fan bracket 25
Reinstall the drive cage 26
Reinstall the motherboard tray 27
Route cables & close the chassis 28
Warranty & support 29
Regulations 30
Manual revision history 31

Exploded assembly & part key

Assembly part key

No.	Part	Related procedure
1	Lid	Remove the lid (page 9)
2	Subfront	Install a front AIO radiator (page 14)
3	Filter foam	Remove the faceplate & filter (page 10)
4	Faceplate	Remove the faceplate & filter (page 10)
5	Front I/O assembly	Front ports & connections (page 8)
6	Rack ear & pull handle	Install rack ears & handles (page 12)
7	Chassis wrap	Main chassis body
8	Removable motherboard tray	Remove the motherboard tray (page 11)

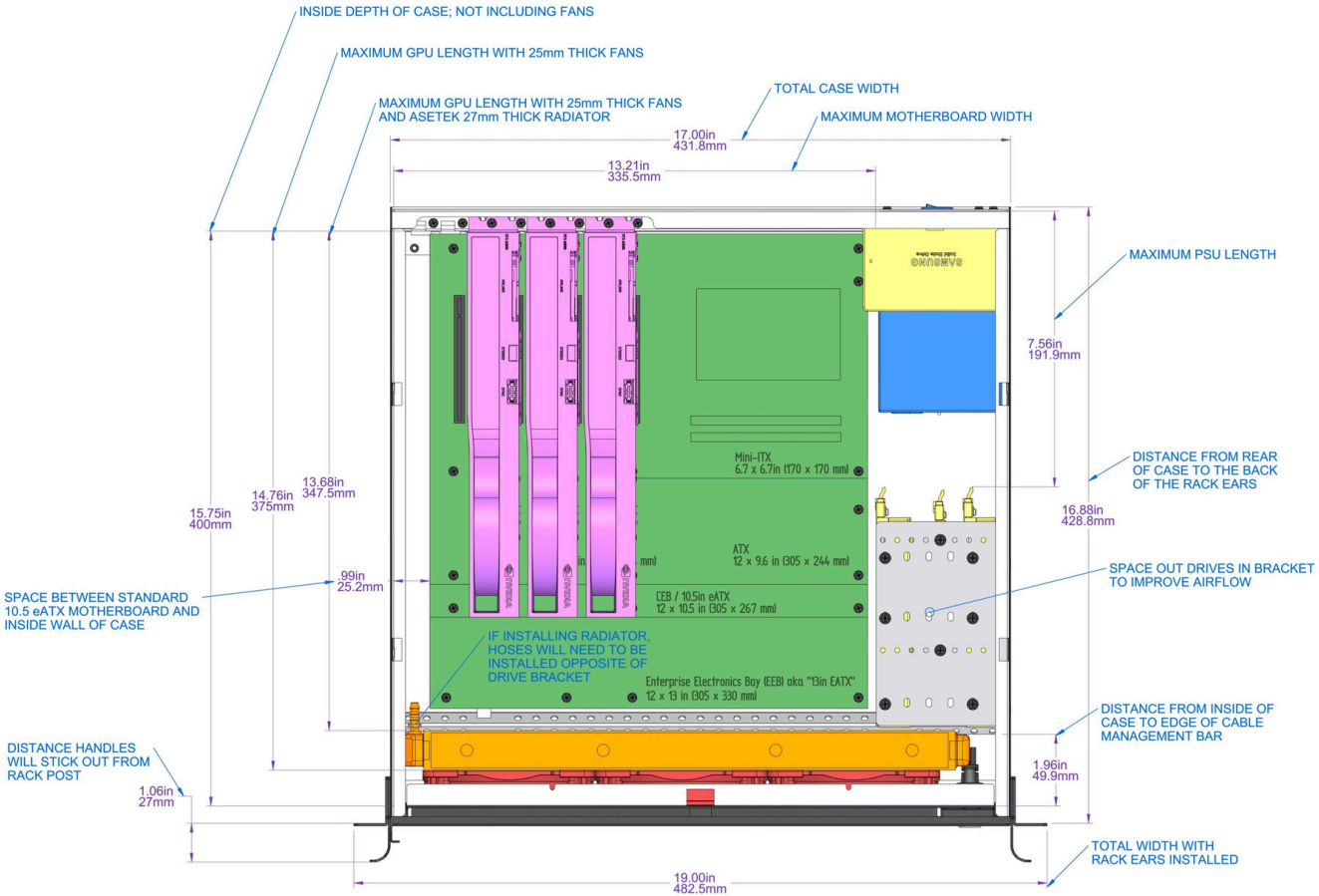
See the hardware-kit page for the screw bag and rack hardware.



Clearance drawing — drive cage installed

This drawing shows the layout with the drive cage installed. Radiator hoses are shown at the opposite end from the cage; drive spacing affects airflow.

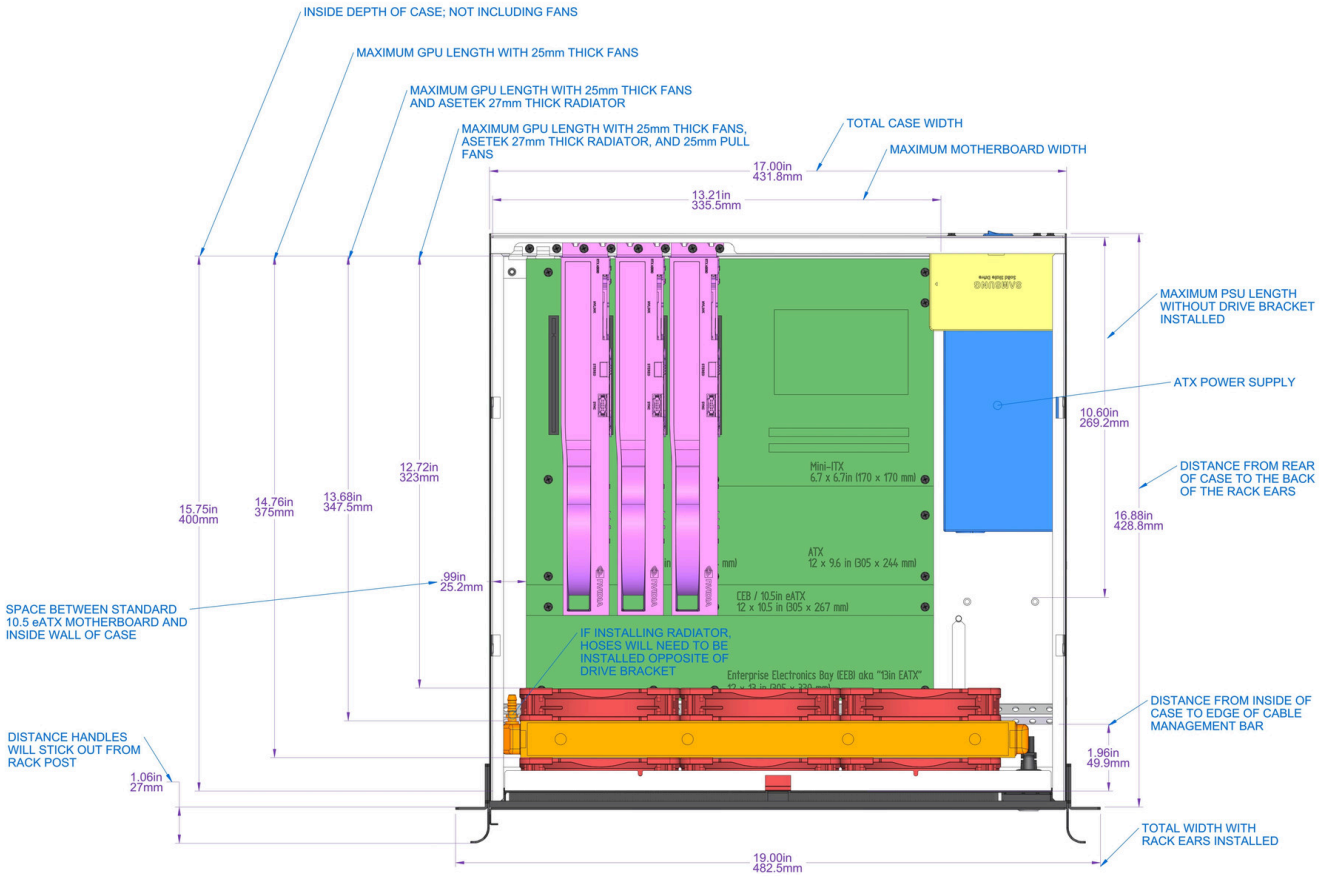
ⓘ Drawing dimensions are nominal. For compatibility, use the limits on the compatibility page.



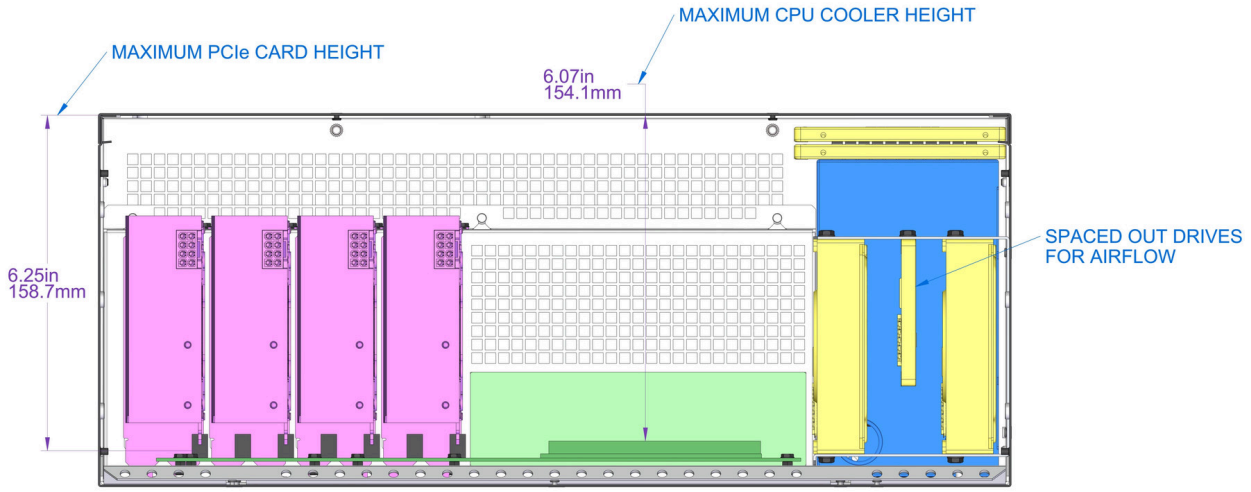
Clearance drawing — cage removed

This drawing distinguishes PSU space **without the drive cage installed** and shows a six-fan push-pull arrangement.

ⓘ Drawing dimensions are nominal. For compatibility, use the limits on the compatibility page.



Front-view component clearances



The front-view drawing illustrates expansion-card height, CPU-cooler height, and spacing between drives for airflow.

ⓘ Drawing dimensions are nominal. For compatibility, use the limits on the compatibility page.

Dimensions & compatibility

Component	Specification
Outer dimensions, D × W × H	431.8 × 431.8 × 176.78 mm
Rack size	4U · 19-inch rack
Power supply	ATX; supported CRPS options require the appropriate mounting arrangement
Motherboard	Mini-ITX, MicroATX, ATX; EATX up to 12 × 10.5 inches; larger boards require dimensional and mounting-pattern checks
CPU air cooler	Up to 153 mm tall
Front cooling	Three 120 mm fans; six with a compatible 360 mm push-pull assembly
Liquid cooling	One 360 mm, one 240 mm, or two 120 mm radiators
Rear cooling	Optional 120 mm rear and PCIe-area fan brackets
Expansion	Eight full-height PCIe slots; cards up to 158 mm tall including power connectors
Storage	Up to 10 × 7 mm 2.5-inch SSDs: eight in the drive cage and two on the motherboard tray (or one 15 mm SSD there; the tray positions are unavailable with the rear fan bracket). The drive cage alternatively holds 3 × 3.5-inch HDDs or mixed HDD/SSD layouts; see drive-cage layouts
Front bays	None

GPU length by cooling configuration

Configuration	Maximum length
No front fans or radiator	395 mm
Three 25 mm front fans	375 mm
Asetek 360 mm AIO with three 25 mm fans	347 mm
Asetek 360 mm AIO with six 25 mm fans (Push-Pull)	322 mm

ⓘ **Six-fan push-pull conflicts with the internal drive cage.** Mounting positions and maximum component counts are not all usable simultaneously.

Hardware kit

Match the **letter and color** here to the references beside the installation illustrations and in each procedure. All rackmount chassis use screwbag **SBAG-HWKIT** (items A–F).

A 6-32 × 0.25-inch hex screws — 45 ×

Yellow-marked · Motherboard, PSU, HDDs, and PCIe accessories.



B M3 × 0.5 screws, 5 mm long — 10 ×

Red-marked · 2.5-inch SSDs and some 3.5-inch accessories.



C 6-32 hex standoffs — 20 ×

Teal in drawings; black, unpatched hardware · 0.125-inch and 0.25-inch lengths. Install in the motherboard tray.



D M3 screws (5 mm thread) — 16 ×

Blue-marked · 2.5-inch SSDs and some 3.5-inch accessories.



E 6-32 × 0.25-inch flat-head screws — 10 ×

Green-marked · Motherboard tray, subfront, internal brackets, and front I/O bracket.



F M4 × 7 mm screws — 7 ×

Pink-marked · Rack ears and pull handles.



G Slot covers — 8 ×

Cover unused PCIe slots.



H Rack ears — 1 pair

Secure the chassis to the rack.



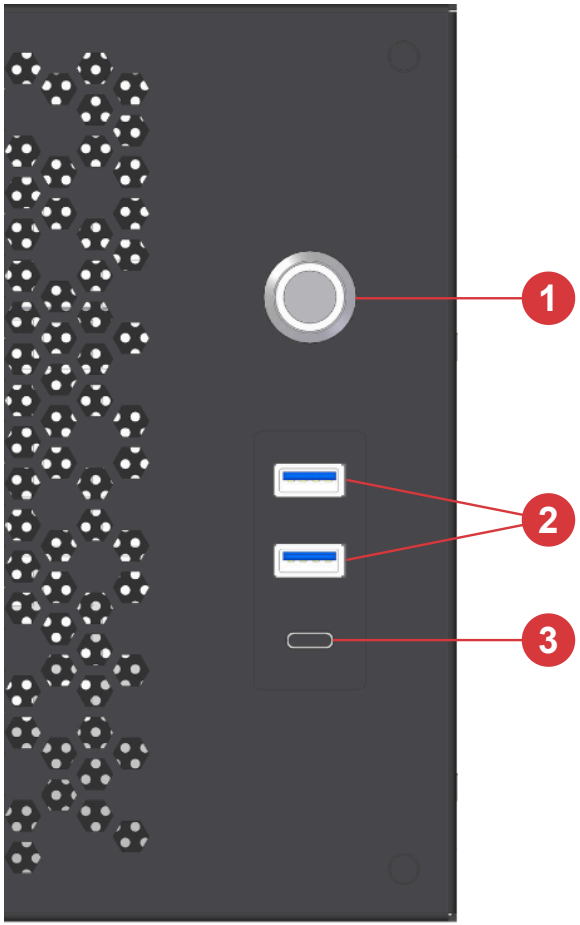
I Pull handles — 1 pair

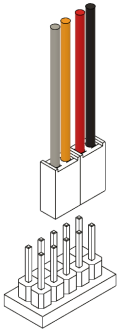
Optional handles make pulling the chassis from the rack easier.



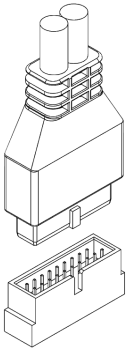
ⓘ Items G–I are separate chassis/accessory items. Illustrations are not to scale. Fan and radiator fasteners are supplied with the cooling components.

Front ports & connections

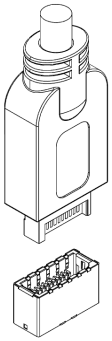




Power switch & status light



USB Type-A ports



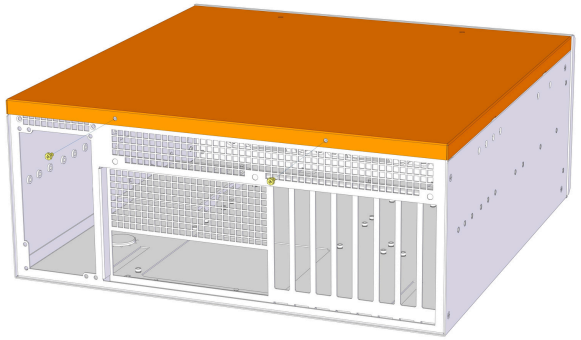
USB Type-C port

No.	Port / connection
1	**Power switch & status light** 16 mm momentary switch with status light. Connect switch and LED leads to the front-panel header; observe the motherboard's LED polarity markings.
2	**USB Type-A ports** Two USB 3.0 Type-A ports, connected through the keyed internal USB cable shown.
3	**USB Type-C port** One USB 3.2 Type-C 10 Gbps port, connected through the internal Type-E cable shown. Requires a matching motherboard header.

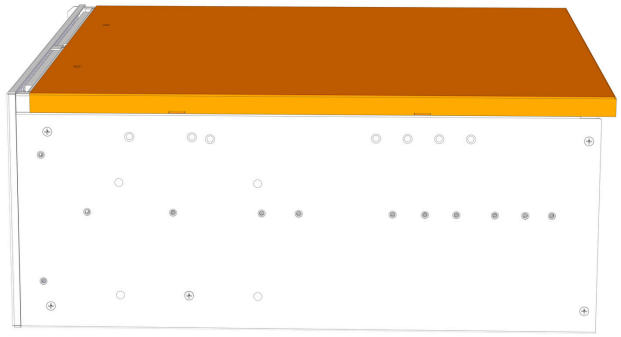
ⓘ Use your motherboard manual to identify matching headers and pin assignments. The numbered front-panel locations correspond to the connector drawings and legend.

Remove the lid

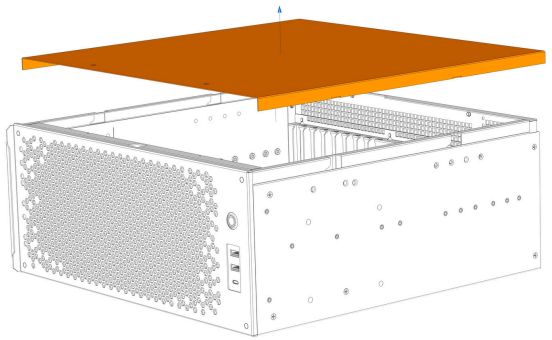
1



2



3



A Yellow

1. Remove the screws

Remove the two yellow-patched 6-32 screws **(A)** along the upper rear edge that secure the lid to the motherboard tray.

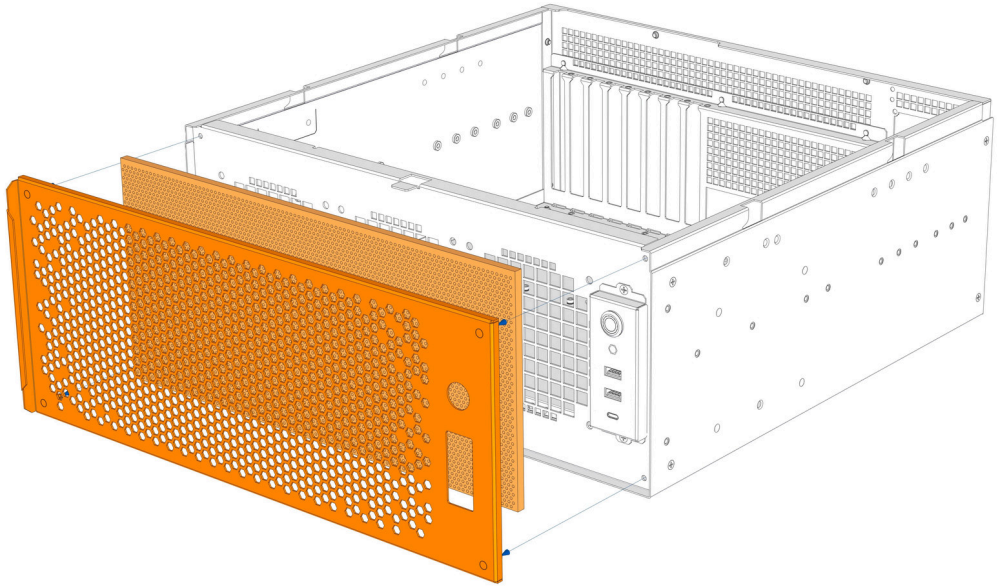
2. Slide the lid back

Push the lid toward the rear of the chassis.

3. Lift the lid away

Check that the lid flanges have cleared the chassis slots. Lift the lid up and away.

Remove the faceplate & filter



1. Release the left side

Pull on the left-side handle of the faceplate until its locating pins disengage, approximately ½ inch.

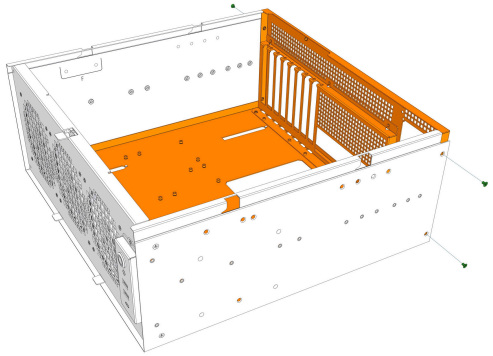
2. Release the right side

Gently pull the right side outward to remove the faceplate and dust filter.

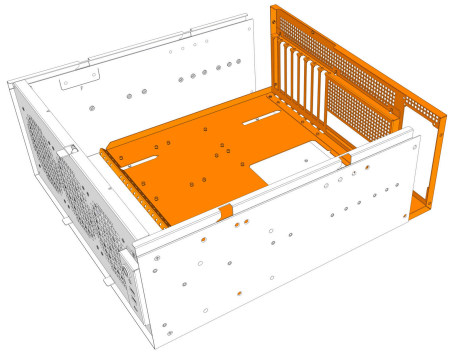
Keep the faceplate and filter aside while installing front fans or a radiator.

Remove the motherboard tray

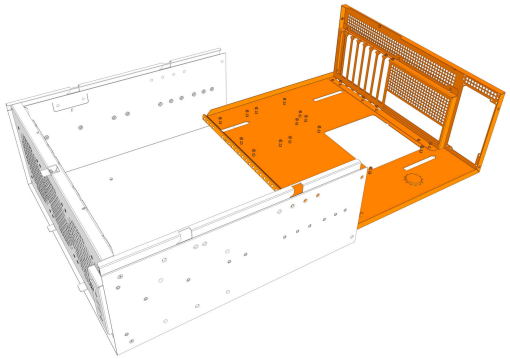
1



2



3



E Green

ⓘ Removing the tray is optional, but can make component installation easier.

1. Remove the tray screws

Remove the four green-patched 6-32 screws (E) securing the tray's rear panel to the chassis, two on each side.

2. Slide the tray back

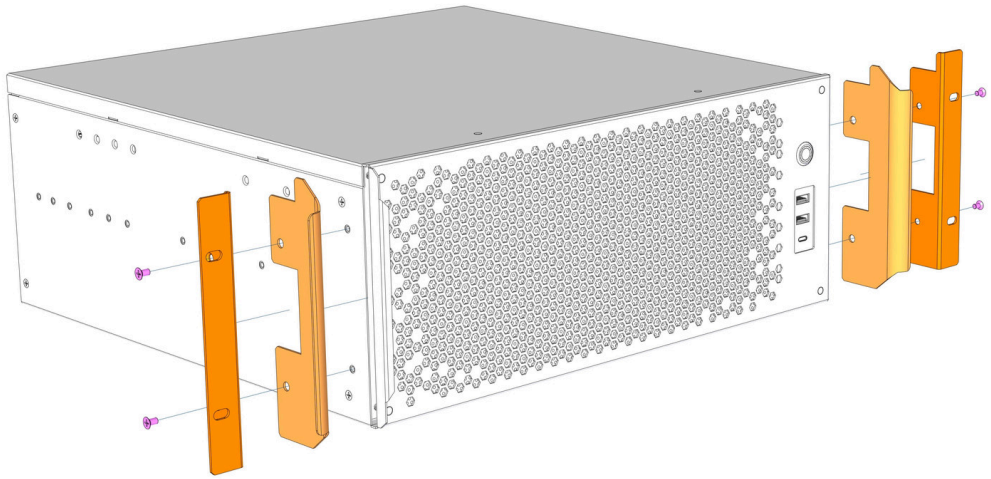
Move the tray toward the rear until all six round metal tabs disengage from the tray.

3. Lift the tray out

Lift only after the tray has cleared the tabs. Do not force it upward while it remains engaged.

Install rack ears & handles

F Pink | H | I



1. Align the hardware

If using pull handles, stack the rack ears **(H)** over the pull handles **(I)** with the handle facing away from the faceplate.

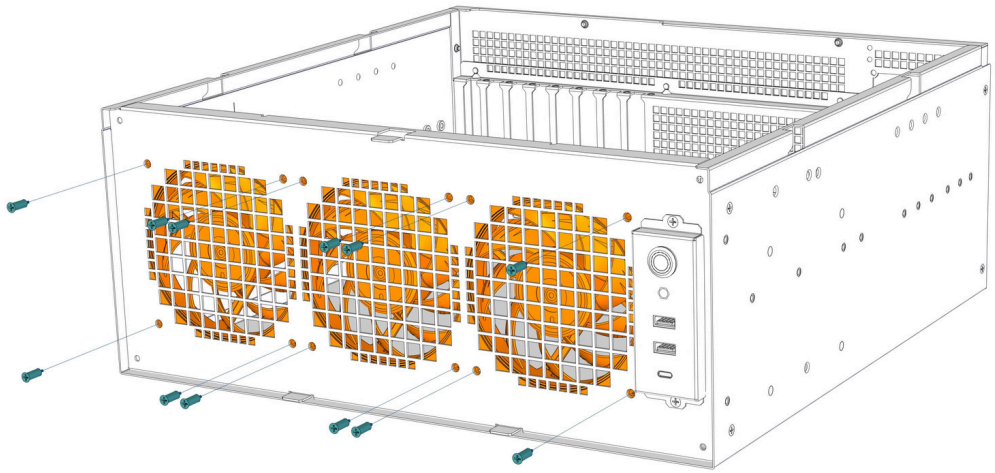
Rack ears can also be installed without handles.

2. Secure both sides

Install 4 pink-marked M4 × 7 mm screws **(F)** in total, 2 on each side, through the rack ears and handles into the chassis. Tighten by hand.

ⓘ Rack-slide installation depends on the rail kit. Follow the instructions supplied with the selected kit.

Install front fans



1. Align the fans

Align three 120 mm fans with the subfront. Each fan uses four mounting points, twelve in total.

2. Secure the fans

Use the self-tapping fan screws supplied with your fans. Pass them through the subfront and into the fan frames.

Tighten sufficiently to prevent movement and vibration.

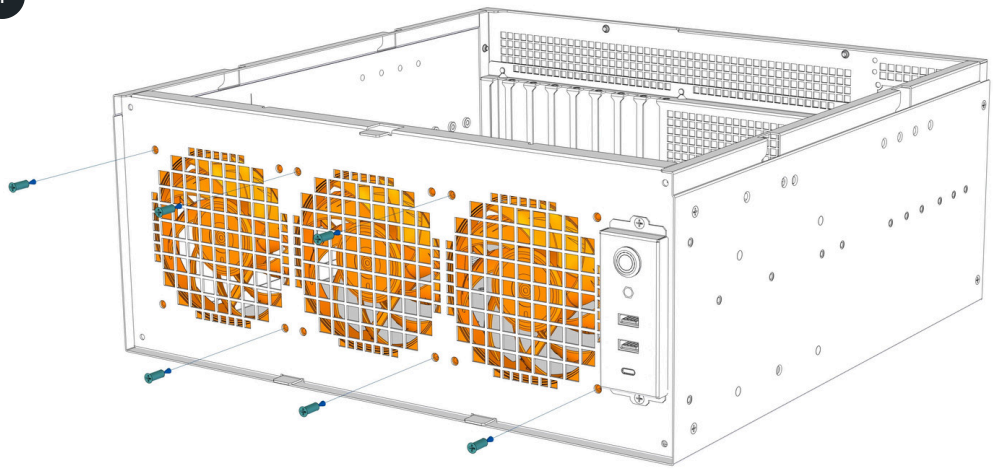
Check GPU clearance

With three 25 mm front fans, maximum GPU length is **375 mm**. Thicker fans reduce available space.

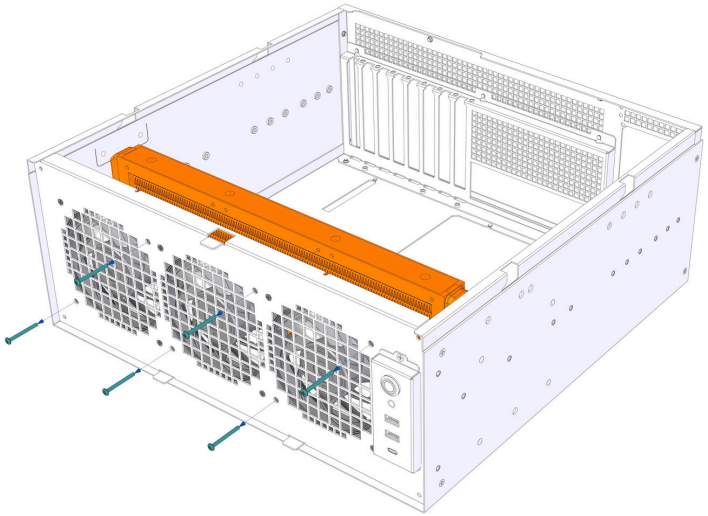
For an AIO configuration, continue to the next section before selecting mounting screws.

Install a front AIO radiator

1



2



1. Position the fans

Place three 120 mm fans between the subfront and the 360 mm radiator. Remove any fan screws occupying the radiator mounting positions.

2. Align and secure the assembly

Align all twelve mounting points in the subfront, fans, and radiator. Use the cooler manufacturer's radiator screws through the subfront and fans into the radiator.

Incorrect screw length can damage the radiator. Check that the radiator clears the front I/O assembly before tightening.

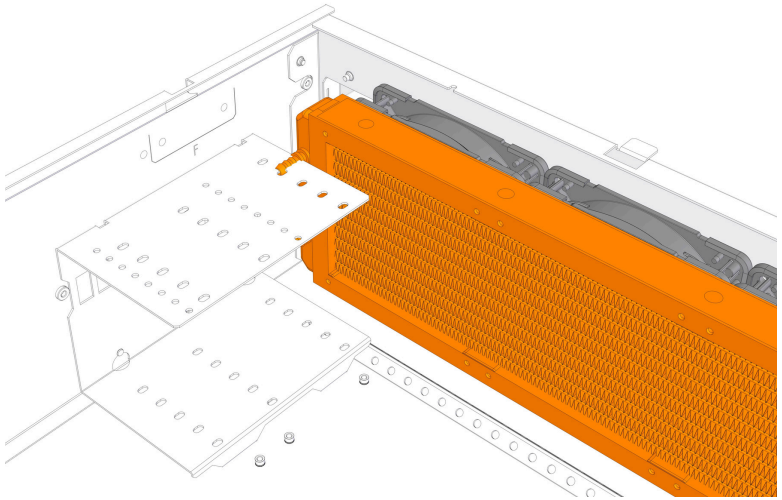
Check component clearance

With the specified Asetek 360 mm AIO and three 25 mm fans, maximum GPU length is **347 mm**.

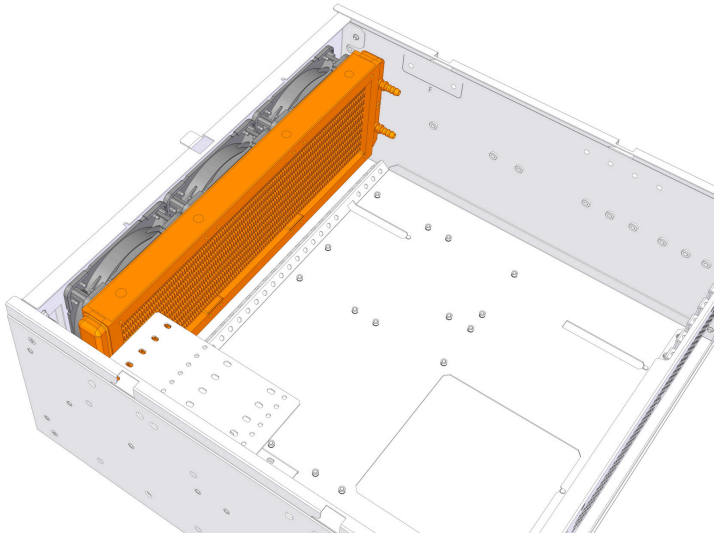
Check the drive-cage and radiator restrictions in Radiator and drive-cage clearance (page 15).

Radiator & drive-cage clearance

1



2



Check the radiator end and tube fittings

The two illustrations show a conflicting arrangement and an arrangement with clearance. Check the radiator end tank, tube fittings, drive cage, and front I/O assembly together before fastening.

Check radiator thickness

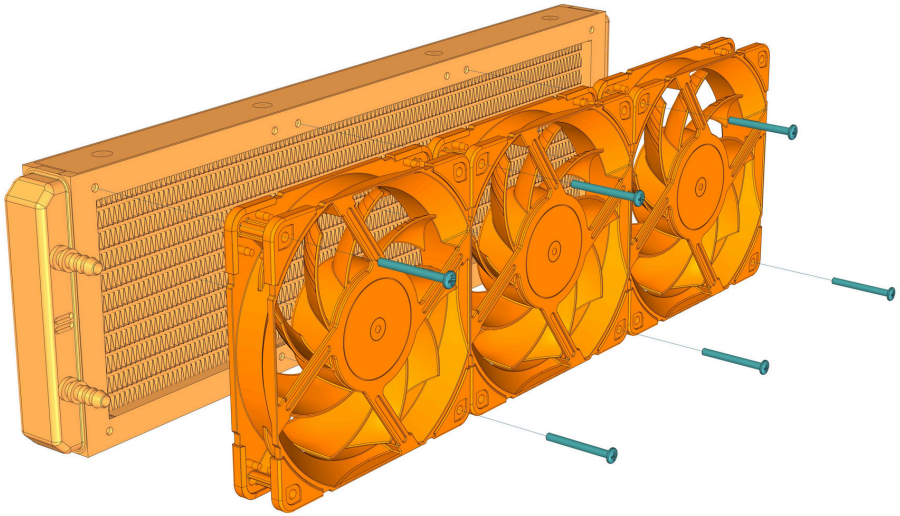
Radiators thicker than 27 mm can interfere with the internal drive cage. Verify the complete fan/radiator thickness and end-tank geometry for the selected cooler.

Preserve the required clearance

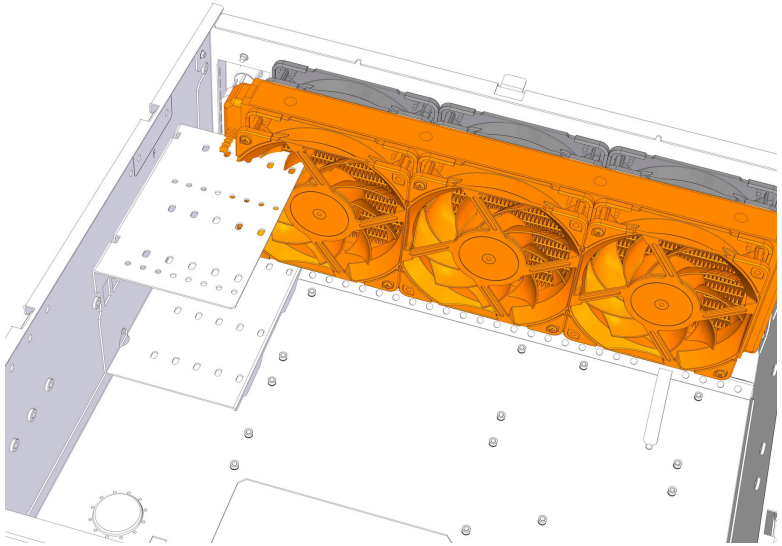
If parts touch or prevent the drive cage from seating, change the configuration or omit the cage. Do not force the cage against the radiator or tubing.

Install push-pull cooling

1



2



1. Check the configuration first

A compatible 360 mm radiator may use three 120 mm fans on each side. **This six-fan arrangement conflicts with the internal drive cage.** Remove the cage before assembling this configuration.

2. Attach the second fan bank

Align the three additional fans with the rear of the radiator. Secure them using the radiator manufacturer's specified screws.

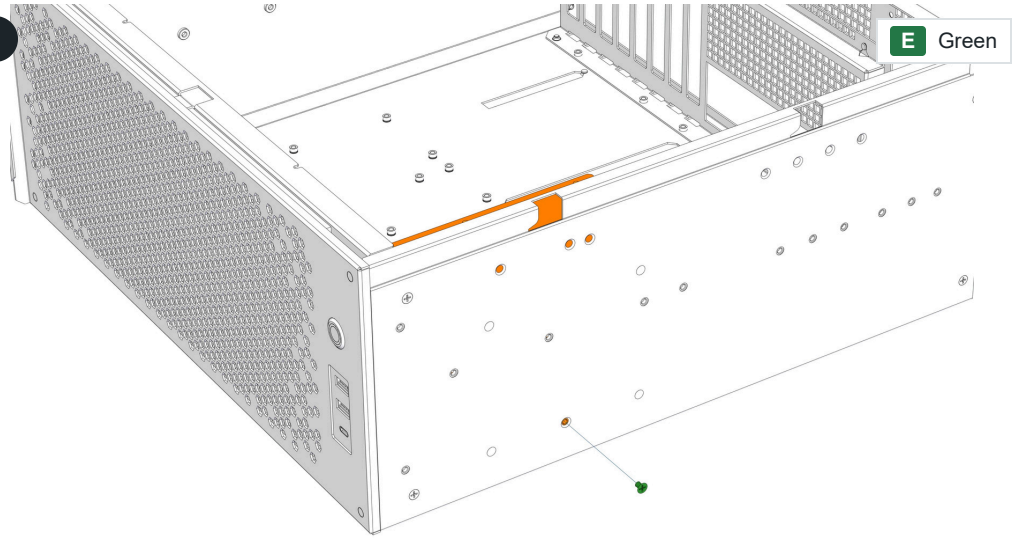
3. Mount the complete assembly

Align the front fan bank, radiator, and subfront. Install the specified screws through the subfront and front fans into the radiator.

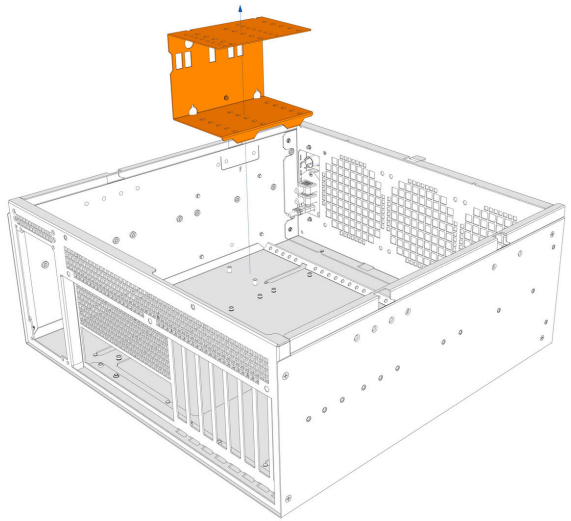
The assembly must clear the motherboard tray, front I/O, and all installed components. With the specified Asetek radiator and six 25 mm fans, maximum GPU length is **322 mm**.

Remove the internal drive cage

1



2



1. Remove the retaining screw

Remove the one green-patched 6-32 screws (E) securing the internal drive cage at the side of the chassis.

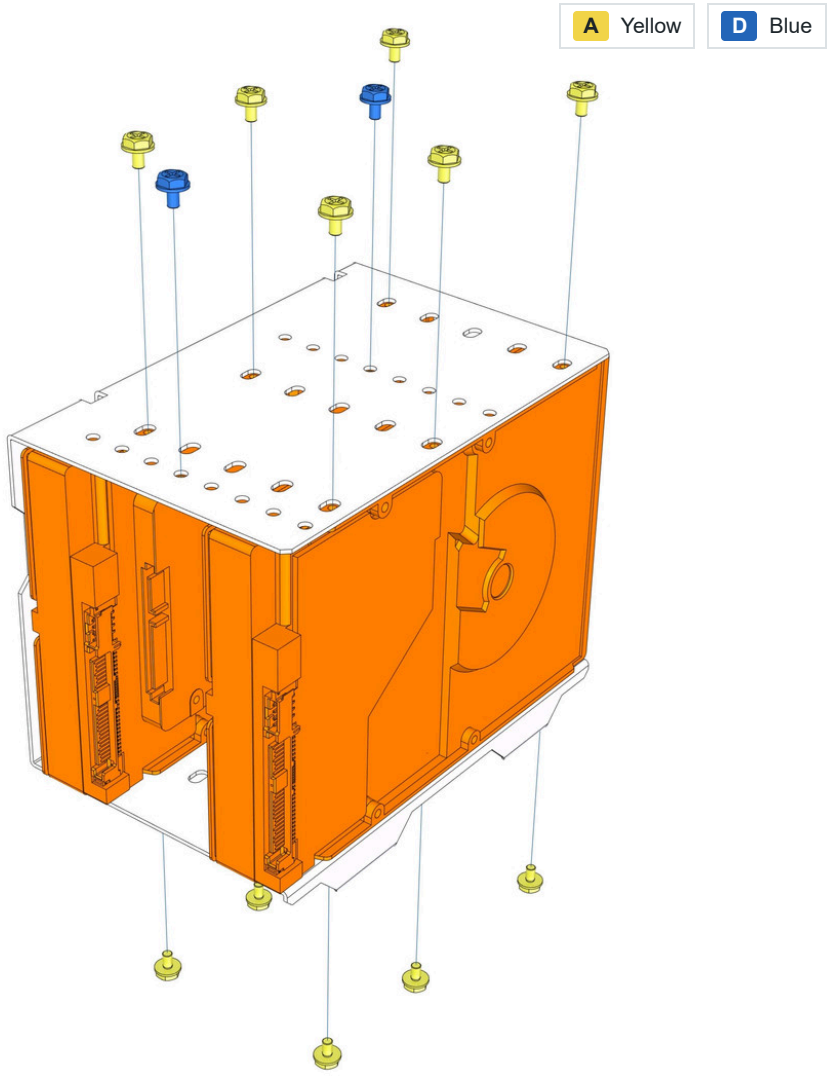
2. Lift the cage out

Slide the cage upward and lift it free of the chassis. Disconnect drive cables first if drives are already installed.

Check the selected cooling layout

The cage occupies space beside the front cooling assembly. Six-fan push-pull is incompatible with the cage. Review Radiator and drive-cage clearance (page 15) before selecting the final drive arrangement.

Install drives in the cage



1. Choose the drive layout

ⓘ The illustration is one populated example, not a promise that every drive mount can be used at the same time. Supported cage layouts include **3 3.5-inch HDDs**, **three HDDs + one 7 mm SSD**, **two HDDs + three 7 mm SSDs**, and **eight 7 mm SSDs**. Nine SSDs fit only when interlocked so that none sit flat.

2. Align the drive mounting holes

Position drives so their connectors can be reached after the cage is installed. Match the cage holes to the drive mounting holes.

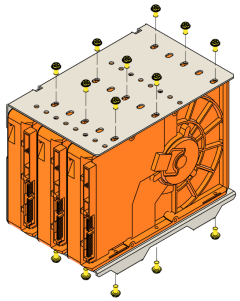
3. Secure the drives

The populated example identifies yellow-patched 6-32 screws **(A)** and blue-patched M3 screws **(D)**. Match the fastener to the drive's thread and permitted screw engagement; do not force mismatched screws.

For alternative populated views, see Drive-cage layout examples (page 19).

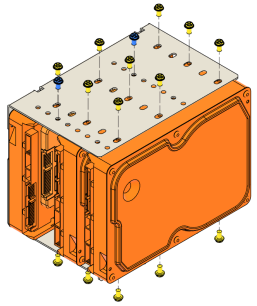
Drive-cage layout examples

ⓘ These are alternative drive-cage layouts. SSD counts below apply to **7 mm-thick 2.5-inch drives**. Check cooler, PSU, and cable clearance for the chosen arrangement; do not add capacities from different layouts.



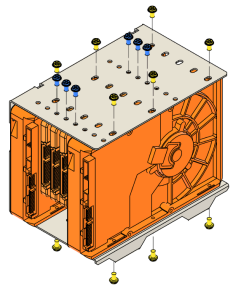
1. Three 3.5-inch HDDs

3 × 3.5-inch HDD.



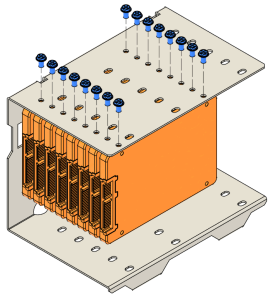
2. Three HDDs + one SSD

3 × 3.5-inch HDD + 1 × 7 mm 2.5-inch SSD.



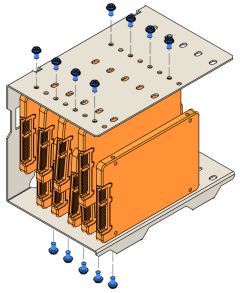
3. Two HDDs + three SSDs

2 × 3.5-inch HDD + 3 × 7 mm 2.5-inch SSD.



4. Eight 2.5-inch SSDs

8 × 7 mm 2.5-inch SSD.

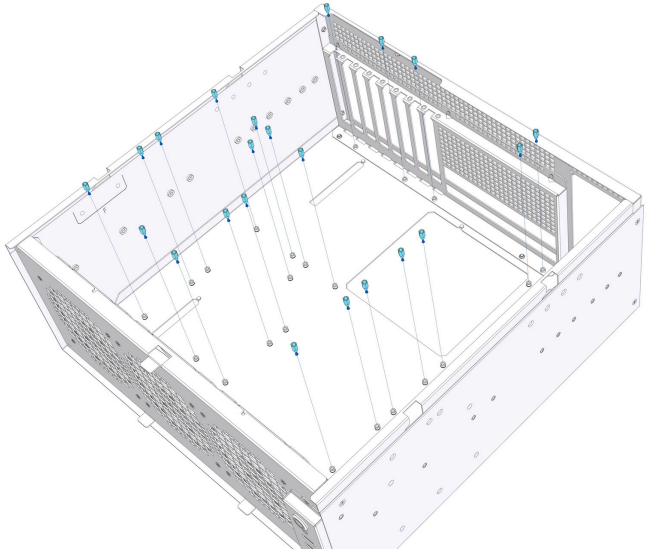


5. Nine 2.5-inch SSDs

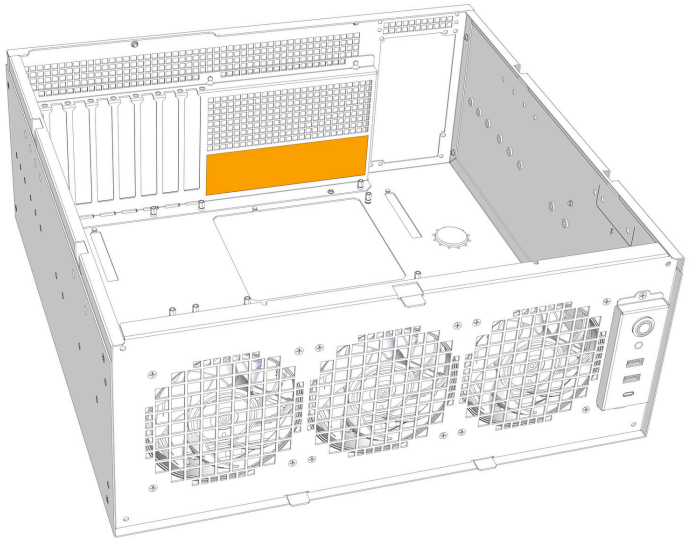
9 × 7 mm 2.5-inch SSD. Fits only with the drives interlocked; none sit flat. Not included in the listed capacity of eight.

Prepare the motherboard tray

1



2



C Teal

1. Install the standoffs

Match standoff positions to the mounting holes on your motherboard. Install the black hex standoffs **(C)** into the corresponding threaded locations on the tray.

Do not leave an extra standoff beneath a part of the board without a mounting hole.

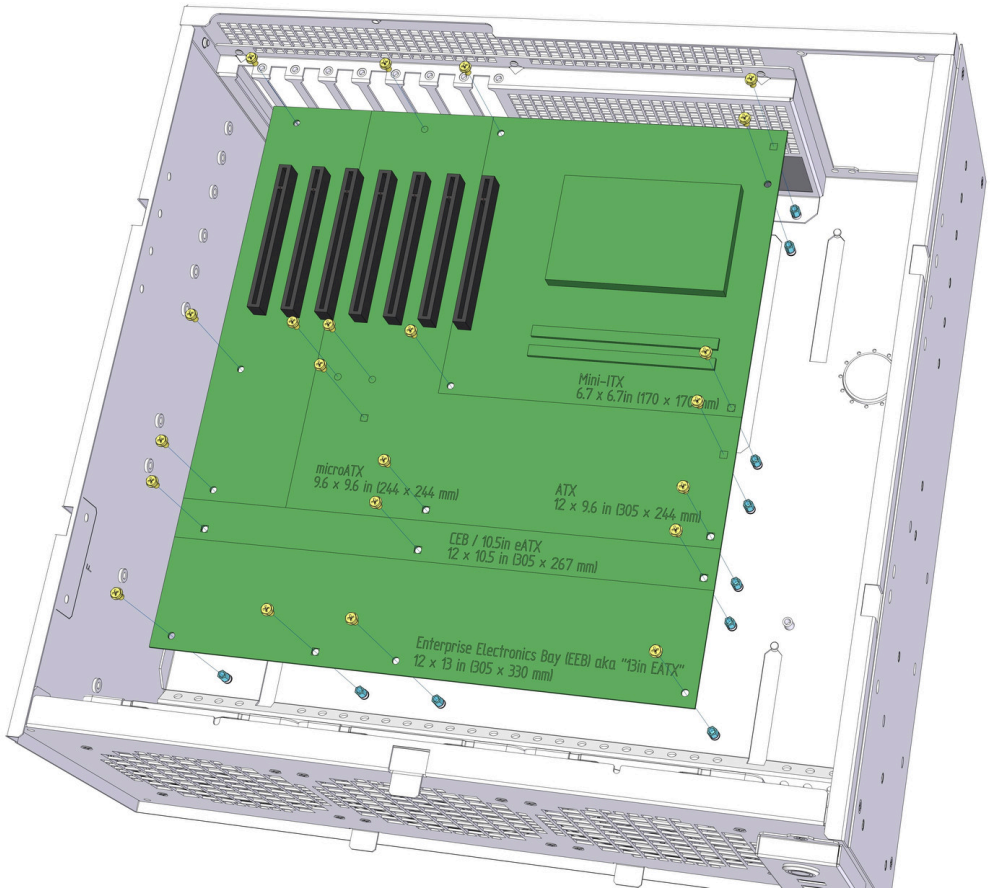
2. Install the I/O shield

If your motherboard has a separate I/O shield, orient it to match the motherboard ports and press it into the rectangular opening in the tray's rear panel.

ⓘ Skip this step if the I/O shield is integrated into the motherboard.

Install the motherboard

A Yellow C Teal



1. Lower the motherboard

Place the motherboard onto the black hex standoffs **(C)**. Align its rear ports with the I/O opening and its mounting holes with the standoffs.

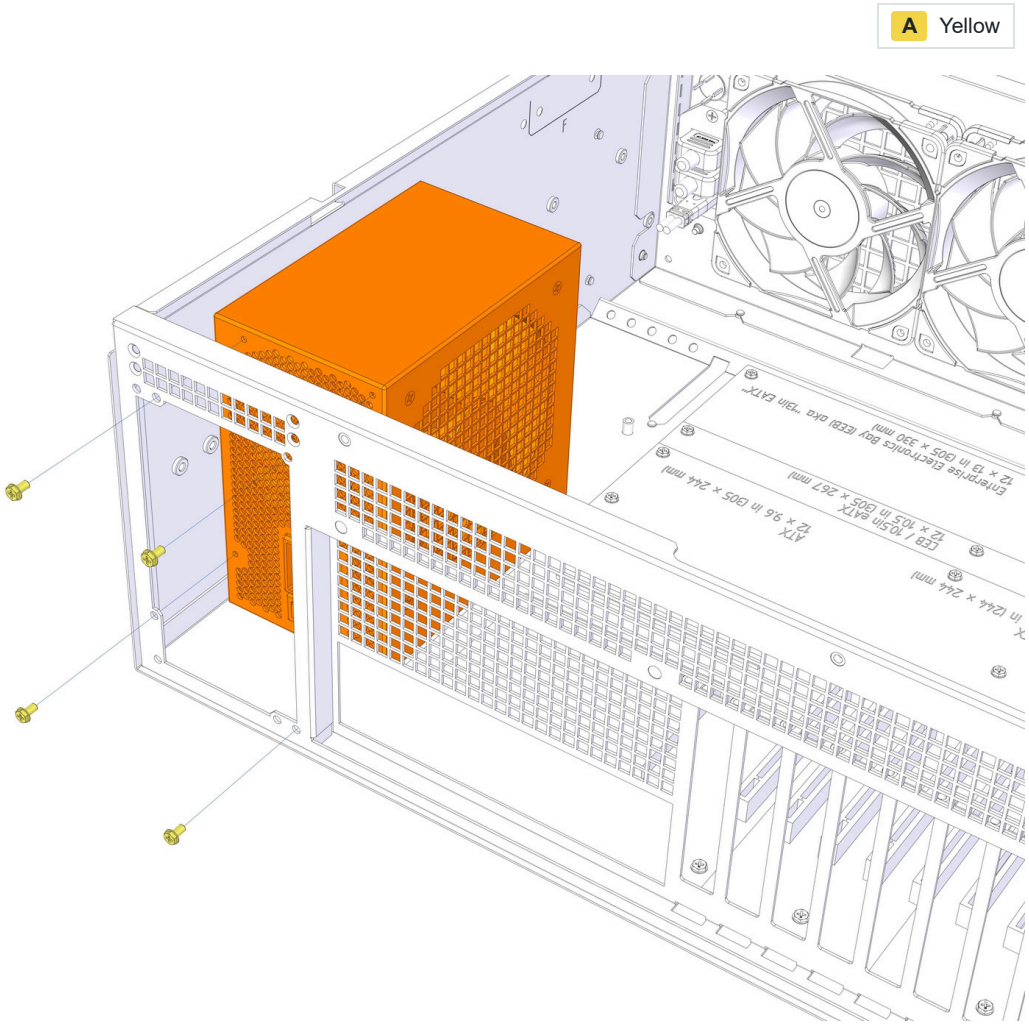
2. Secure the board

Install yellow-patched 6-32 screws **(A)** through the board and into the standoffs.

Tighten by hand to avoid damaging the board or loosening the standoffs during future removal.

Follow your motherboard's instructions for CPU, memory, M.2 drives, and cable connections.

Install the power supply



1. Position the ATX power supply

Place the ATX PSU into the mounting location shown on the motherboard tray. Align its rear mounting holes and keep the fan intake clear.

2. Secure the PSU

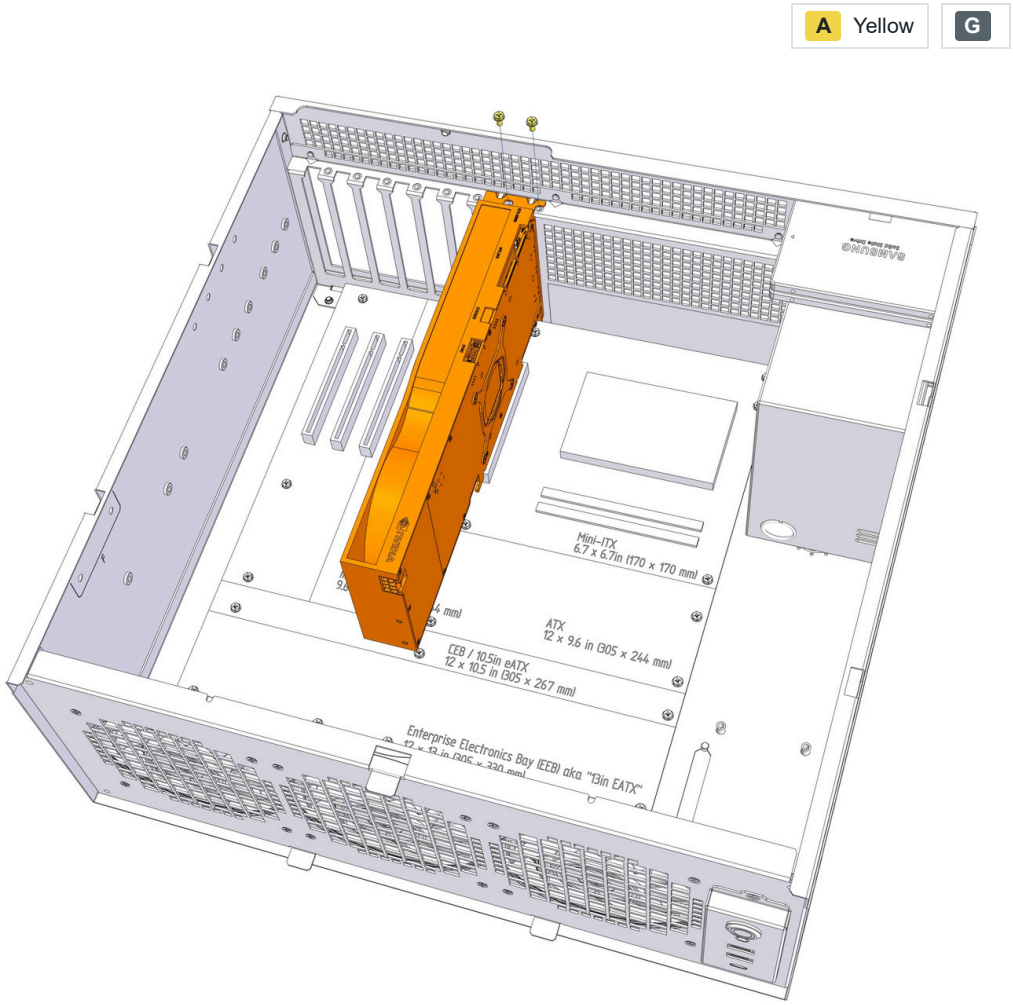
Install yellow-patched 6-32 screws **(A)** through the tray into the PSU. Tighten by hand and route cables clear of tray engagement points.

Check length and cable clearance

The maximum PSU length is **190 mm including connectors**, with or without the drive cage.

ⓘ **CRPS options:** CRPS mounting depends on the PSU format and adapter. This page illustrates ATX installation only; obtain instructions for the exact CRPS adapter before installation.

Install a GPU or PCIe card



1. Prepare the slot

Select the PCIe slot specified by your motherboard manual. Remove the corresponding chassis slot covers and open the motherboard slot latch, if present.

2. Seat the card

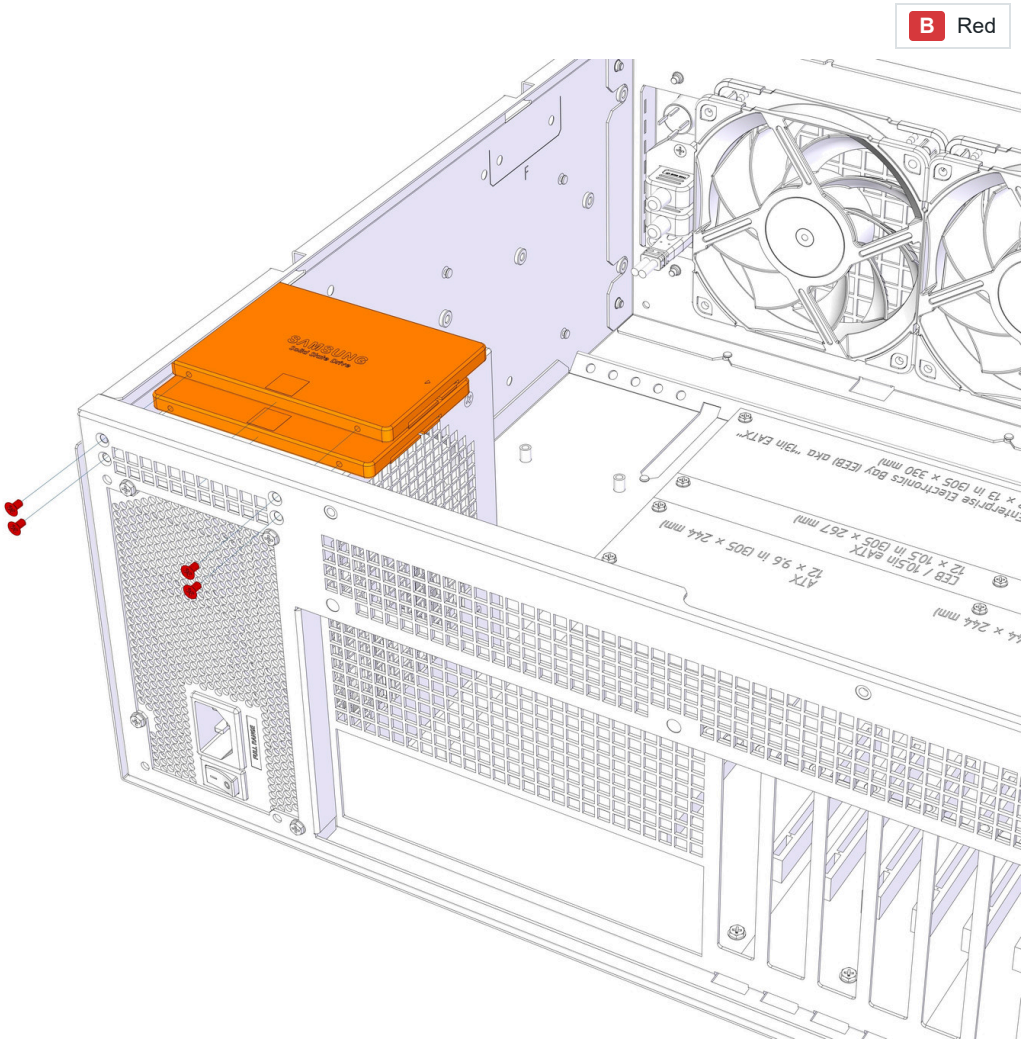
Align the card with the motherboard slot and rear bracket opening. Lower it carefully and apply gentle, even pressure until fully seated.

3. Secure the bracket

Fasten the card's mounting bracket to the tray with yellow-patched 6-32 screws **(A)**. Fit slot covers **(G)** over unused openings and secure them with yellow-patched 6-32 screws **(A)**.

ⓘ Card height, including power connectors, must remain within **158 mm**. Check the cooling-dependent length limits before installation.

Install a 2.5-inch SSD



Rear fan bracket clearance: If the optional rear fan bracket is installed, neither SSD mounting position on the tray can be used. This restriction applies to all 4U chassis.

1. Position the SSD

The rear of the motherboard tray has two pairs of 2.5-inch mounting holes. Install either two SSDs up to 7 mm thick, or one SSD up to 15 mm thick. Align the drive with the mounting holes shown. Keep its connectors accessible and check clearance to the PSU and motherboard.

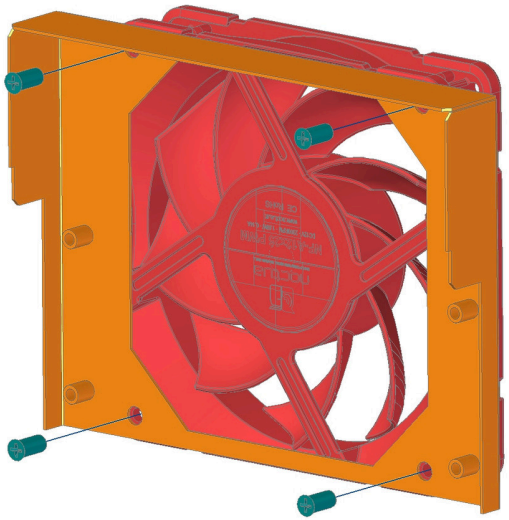
2. Secure the drives

Secure each drive with two red-patched M3 × 0.5 screws, 5 mm long (B) through the tray. Two 7 mm drives use four screws in total. Tighten by hand without forcing the threads.

ⓘ For M.2 storage, follow the motherboard's installation instructions.

Install the rear fan bracket

1



2



An optional rear fan bracket holds one 120 mm fan behind the rear vent of the chassis. It is available separately as KIT-4U-REARFAN-B (bracket) or KIT-4U-REARFAN-NF25 (bracket with a Noctua NF-A12x25 fan). Every 4U chassis uses the same bracket and steps.

1. Mount the fan on the bracket

Secure a 120 mm fan to the bracket with the four fan screws supplied with the kit, or with the screws supplied with the fan. Orient the fan to exhaust air out of the chassis.

2. Fit the bracket

Hold the bracket against the inside of the rear panel, behind the vent. From outside the chassis, secure it through the rear panel with the four screws supplied with the kit.

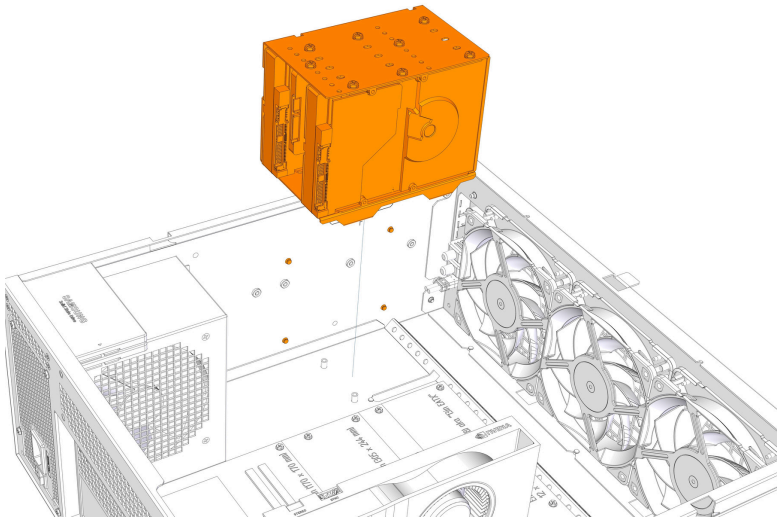
3. Connect the fan

Connect the fan to a motherboard fan header and route the cable clear of the CPU cooler and fan blades.

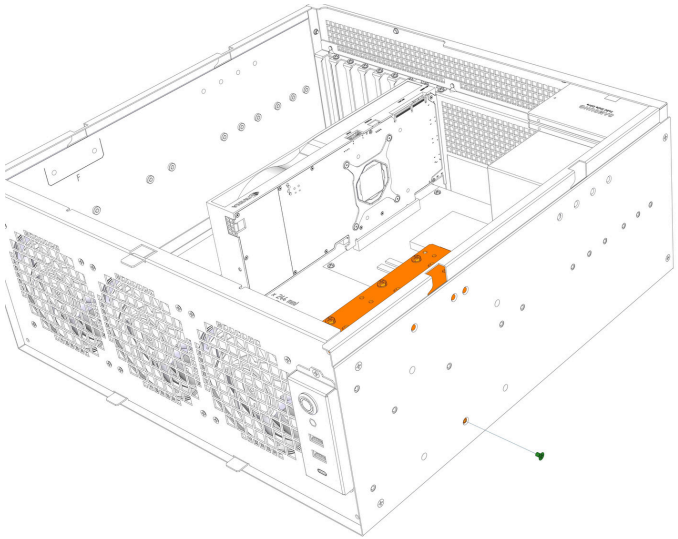
ⓘ The rear 2.5-inch SSD positions on the motherboard tray cannot be used while the rear fan bracket is fitted.

Reinstall the drive cage

1



2



E Green

1. Check cooler clearance

Verify that the selected drives, radiator, and fan configuration allow the cage to seat fully. Do not install the cage with six-fan push-pull cooling.

2. Seat the cage

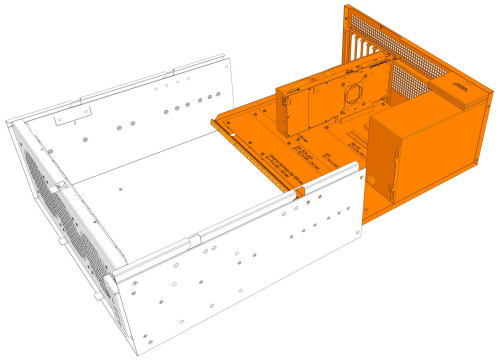
Lower the populated cage into its chassis mounting position, aligning its engagement points.

3. Secure the retaining screw

Reinstall the green-patched 6-32 screws (E) through the side of the chassis into the cage. Connect the drive cables without obstructing fans or the removable motherboard tray.

Reinstall the motherboard tray

1

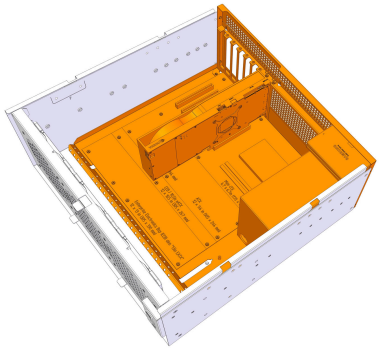


E Green

1. Place and slide the tray

Lower the assembled tray into the chassis and slide it forward. Lift gently as needed to prevent its front edge catching on the chassis tabs.

2



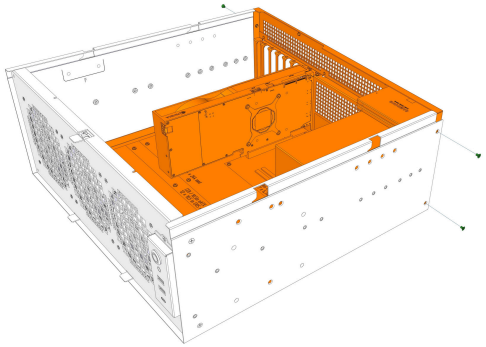
2. Engage all six tabs

Continue forward until all six tabs are seated in the tray and its rear panel sits flush against the chassis rear.

3. Secure the tray

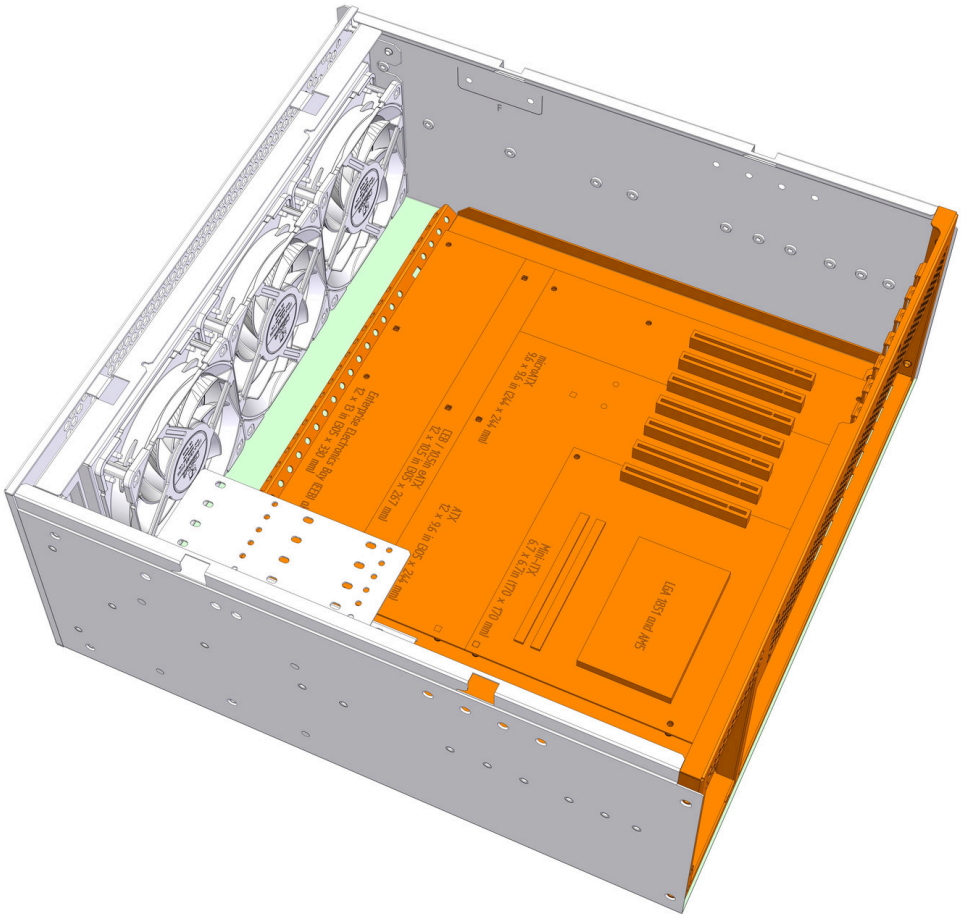
Install four green-patched 6-32 screws (E), two on each side, through the chassis into the tray. Tighten by hand.

3



E Green

Route cables & close the chassis



1. Route and secure cables

Route front-panel, PCIe, power, and fan cables along the front edge of the motherboard tray's cable-management bar, highlighted in green. Bundle excess slack behind the PSU and secure it with ties.

Keep all cables clear of fan blades, drives, and the tray's engagement points. Release ties that anchor cables to the fixed chassis before removing the tray.

2. Close the chassis

Refit the filter and faceplate, aligning their locating pins. Align the lid flanges with the chassis slots, lower the lid, and slide it forward. Secure the lid with its two yellow-patched 6-32 screws **(A)**.

Warranty & support

WARRANTY POLICY

Sliger Designs LLC ("Sliger") warrants its Products, if properly used and installed, against defects in material and workmanship for the period stated on the product specifications. Under normal use and service, every hardware portion of the products will be free from physical defects in material and workmanship during the warranty period (excludes normal wear and tear), or the product will be repaired or replaced as determined solely by Sliger.

For full warranty and RMA policy visit :
https://www.sliger.com/support/warranty_and_rma/

SUPPORT AND SERVICE

If you have any questions or problems with the Sliger product you purchased, please don't hesitate to contact us via email or phone. Please include a detailed explanation of your problem and your order number.

For comments and suggestions, you can e-mail our design team, orders@sliger.com.

Lastly we would like to thank you for your support by purchasing this product.

For more information about Sliger Designs, please visit us online.

Website: sliger.com
Support Email: orders@sliger.com
Phone Number: 1 775 356 5595
Hours of Operation: 8a - 4:30p MST

Regulations

To the best of our knowledge, all computer chassis manufactured by Sliger Designs, LLC adhere to ROHS, REACH, and Conflict Minerals regulations. All Sliger products are produced in the United States of America and primarily utilize materials sourced domestically.

ROHS STATEMENT

This product is compliant to implemented RoHS (Restriction of Hazardous Substances) type regulations worldwide, including but not limited to China, India, Japan, Korea and European Union.

REACH STATEMENT

REACH (Registration, Evaluation, Authorization and Restriction of Chemicals, EC 1907/2006) is the European Union's (EU) chemical substances regulatory framework. REACH requires Sliger Designs, LLC to provide customers with sufficient information on Substances of Very High Concern (SVHC) contained in products in concentration above 0.1% weight by weight (w/w) to allow safe use of the product.

Certificates of Compliance are available upon request.

Manual revision history

Version	Date	Changes
v1.0	2026-11-09	Initial release
v1.1	2026-11-13	Fan and AIO steps: the front fan mount is called the subfront throughout.